

PREDICTING CUSTOMER CHURN: A DATA-DRIVEN ANALYSIS OF SUBSCRIPTION CUSTOMER BEHAVIOUR

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DECLARATION

I, Qurat Ul Ain, hereby declare that this capstone project report titled "Predicting Customer Churn: A Data-Driven Analysis of Subscription Customer Behaviour" is my own original work, carried out as part of the Data Science & AI Training Program conducted by the India Data Research Academy (IDRA). All the analysis, code and interpretations presented in this report were completed by me. Any external dataset, tool, or reference material used has been duly acknowledged in the References section. This work has not been submitted elsewhere for any other assessment.

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ABSTRACT

Customer churn — the loss of subscribers to a competing or discontinued service — directly erodes recurring revenue for subscription-based businesses. This capstone project analyses the IBM Telco Customer Churn dataset (7,043 customers, 21 attributes) to understand which customer characteristics are associated with churn and to build a predictive model that flags at-risk customers. After cleaning the dataset (correcting 11 non-numeric TotalCharges values and standardising redundant category labels), exploratory data analysis showed that customers on month-to-month contracts, with shorter tenure and higher monthly charges, churn at a substantially higher rate than customers on annual contracts with longer tenure. Three engineered features — tenure group, average monthly spend and number of subscribed services — were added to support the analysis. Two classification models, Logistic Regression and Random Forest, were trained on an 80/20 stratified train-test split with one-hot encoded categorical features and standardised numerical features. Logistic Regression was selected as the primary model, achieving 79.9% test accuracy, 65.3% precision, 51.9% recall, an F1-score of 57.8%, and a ROC-AUC of 0.843. Contract type, tenure, total charges and internet service type were the most influential predictors. The findings indicate that early-tenure, month-to-month customers on fiber-optic plans represent the highest-risk segment, and the report concludes with targeted retention recommendations for this segment.

Keywords: customer churn; exploratory data analysis; classification; logistic regression; random forest; predictive analytics

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1. INTRODUCTION

1.1 Background

Subscription-based businesses — telecommunications, streaming, SaaS and utilities — depend on recurring revenue from an installed customer base. Customer churn, the rate at which customers discontinue their subscription, is one of the most closely watched metrics in these industries because acquiring a new customer typically costs several times more than retaining an existing one. Understanding why customers leave, and identifying which customers are at risk before they leave, allows a business to act proactively rather than reactively.

1.2 Context of the Problem

This project uses the IBM Telco Customer Churn dataset, a widely used benchmark dataset representing a telecommunications company offering phone and internet services. Each row represents one customer, with fields describing demographics, subscribed services, contract and billing details, and whether the customer churned in the last month. A retention team at such a company would use a churn prediction model to prioritise outreach, design targeted offers, and measure the effectiveness of retention campaigns.

1.3 Scope of the Study

The project covers data cleaning, exploratory data analysis, statistical summarisation, feature engineering, and the training and evaluation of two classification models (Logistic Regression and Random Forest) to predict churn. The project does not cover live deployment of the model, real-time scoring infrastructure, or a causal/experimental analysis of retention interventions — these are noted as future work.

2. PROBLEM STATEMENT AND MOTIVATION

Can a customer's demographic profile, subscribed services, contract type and billing history be used to predict whether they will churn within the next billing cycle? Roughly 26.5% of customers in this dataset have churned, representing a significant and costly loss of recurring revenue. A reliable, interpretable classification model would let the business identify high-risk customers early and target retention efforts (discounts, contract upgrades, service improvements) where they are most likely to be effective, rather than applying costly blanket retention campaigns to the entire customer base.

3. OBJECTIVES

- To understand and clean the Telco Customer Churn dataset for analysis.
- To identify the customer characteristics and service patterns most strongly associated with churn.
- To engineer features (tenure group, average monthly spend, service count) that better capture customer behaviour.
- To build a classification model to predict whether a customer will churn.
- To evaluate the model using accuracy, precision, recall, F1-score and ROC-AUC, and compare it against a baseline.
- To translate the results into actionable, evidence-based retention recommendations.

4. RESEARCH QUESTIONS

1. What are the major characteristics and data quality issues in the Telco Customer Churn dataset?
2. Which variables (contract type, tenure, charges, services) are most strongly associated with churn?
3. How does churn vary across contract types, internet service types and tenure groups?
4. How accurately can a classification model predict customer churn, and which model performs best?
5. Which features contribute most to the model's predictions, and are they consistent with the EDA findings?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The dataset used is the IBM Telco Customer Churn dataset, a publicly available benchmark dataset for churn analysis, accessed on 19 September 2026 via a public GitHub dataset mirror (plotly/datasets). It is provided under an open, freely redistributable licence for educational and analytical use.

5.2 Dataset Size and Structure

The raw dataset contains 7,043 customer records and 21 columns, covering demographics, account information, subscribed services, billing details and the churn outcome. After cleaning and feature engineering (Section 6-7), the working dataset has 22 original columns plus 3 engineered features. The target variable is Churn (Yes/No), indicating whether the customer left within the last month.

Data Type	Count of Columns
str	18
int64	2
float64	2

Table 1. Summary of Dataset Column Data Types

5.3 Data Dictionary

Column 1 (Variable)	Column 2 (Meaning)	Column 3 (Data Type)	Column 4 (Role)
customerID	Unique customer identifier	Categorical	Identifier (dropped for modelling)
gender	Customer gender	Categorical	Feature
SeniorCitizen	Whether customer is a senior citizen	Categorical	Feature
Partner	Whether customer has a partner	Categorical	Feature
Dependents	Whether customer has dependents	Categorical	Feature
tenure	Number of months the customer has stayed	Numerical	Feature
PhoneService / MultipleLines	Phone service subscription details	Categorical	Feature

Column 1 (Variable)	Column 2 (Meaning)	Column 3 (Data Type)	Column 4 (Role)
InternetService	Type of internet service (DSL/Fiber/None)	Categorical	Feature
OnlineSecurity...StreamingMovies	Add-on service subscriptions	Categorical	Feature
Contract	Contract term (month-to-month/one year/two year)	Categorical	Feature
PaperlessBilling	Whether billing is paperless	Categorical	Feature
PaymentMethod	Method of payment	Categorical	Feature
MonthlyCharges	Current monthly charge amount	Numerical	Feature
TotalCharges	Total amount charged to the customer	Numerical	Feature
Churn	Whether the customer left within the last month	Categorical	Target

Table 2. Data Dictionary (Key Variables)

6. DATA CLEANING

The raw dataset was largely well-structured, but three issues required attention before analysis: a numeric column stored as text with blank entries, category labels that duplicated information already captured elsewhere, and a categorical flag encoded as 0/1 instead of readable labels.

6.1 Missing Values

The TotalCharges column was stored as text and contained 11 blank-string values (0.16% of records), all belonging to customers with tenure = 0 (i.e., customers who had not yet completed a billing cycle). These were coerced to numeric and the resulting missing values were filled with 0, which correctly reflects that no charges had yet accrued, rather than being dropped or filled with an arbitrary average that would misrepresent new customers.

6.2 Duplicate Records

No duplicate customerID values were found in the dataset (0 duplicates), so no records were removed on this basis.

6.3 Data Types and Inconsistent Values

The TotalCharges column was converted from object (text) to float64. The SeniorCitizen flag was converted from numeric 0/1 to Yes/No for consistency with the other Yes/No demographic columns. Six add-on service columns (OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies) and MultipleLines contained a redundant “No internet/phone service” category that duplicates information already present in the InternetService and PhoneService columns; these were standardised to “No” to avoid an inflated, redundant category level during one-hot encoding.

6.4 Outlier Handling

Numerical variables (tenure, MonthlyCharges, TotalCharges) were checked using boxplots and the IQR method (Table 4). No implausible values were found — tenure ranges from 0 to 72 months (a 6-year observation window) and charges are within a realistic commercial range for a telecom provider — so no observations were removed as outliers. Retaining the full range was considered appropriate because extreme-but-valid values (e.g., long-tenure, high-spend customers) are informative for the churn model rather than being data-entry errors.

6.5 Before vs After Cleaning

Check	Before Cleaning	After Cleaning
Rows	7043	7043
Columns	21	22
Missing values (TotalCharges)	11	0
Duplicate customer IDs	0	0
Non-numeric TotalCharges	11	0

Table 3. Data Quality Before and After Cleaning

7. DATA PREPROCESSING

7.1 Categorical Encoding

All categorical features (gender, SeniorCitizen, Partner, Dependents, service subscriptions, Contract, PaperlessBilling, PaymentMethod, TenureGroup) were one-hot encoded using pandas `get_dummies` with `drop_first=True`, avoiding the dummy-variable trap while keeping the encoding simple and interpretable for the Logistic Regression baseline.

7.2 Feature Scaling and Transformation

The numerical features (tenure, MonthlyCharges, TotalCharges, AvgMonthlySpend, NumServices) were standardised using `StandardScaler` (zero mean, unit variance). Scaling is important for Logistic Regression, whose coefficients and convergence are sensitive to feature magnitude; it has no effect on the Random Forest's splits but was applied consistently across both models for a fair comparison.

7.3 Train-Test Split

The data was split 80/20 into training and test sets (5,634 training / 1,409 test records) using stratified sampling on the target variable to preserve the 26.5% churn rate in both splits, with `random_state=42` for reproducibility. The `StandardScaler` was fit only on the training split and then applied to the test split, preventing information from the test set from leaking into the scaling parameters.

7.4 Final Feature Set

The final model used 28 input features after encoding (demographic flags, service subscriptions, contract and billing details, and the three engineered features) to predict the binary target `Churn_Flag` (1 = churned, 0 = retained).

8. EXPLORATORY DATA ANALYSIS

8.1 Univariate Analysis

```
sns.histplot(df['tenure'], bins=30, kde=True)
plt.title('Distribution of Customer Tenure (Months)')
```

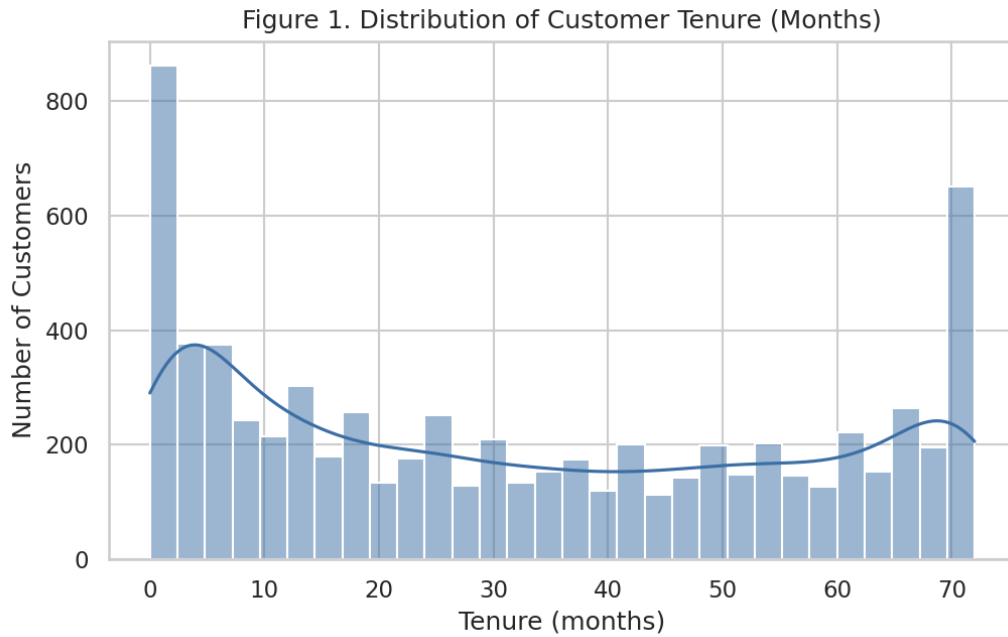


Figure 1. Distribution of Customer Tenure (Months)

Tenure is heavily right-skewed with a large spike near 0-2 months and a smaller secondary peak near 70-72 months. This bimodal-like pattern suggests two distinct populations: customers who churn quickly after joining, and a loyal long-tenure group that stays for the full observation window.

8.2 Categorical Analysis

```
ct = pd.crosstab(df['Contract'], df['Churn'], normalize='index') * 100
ct.plot(kind='bar', stacked=True)
```

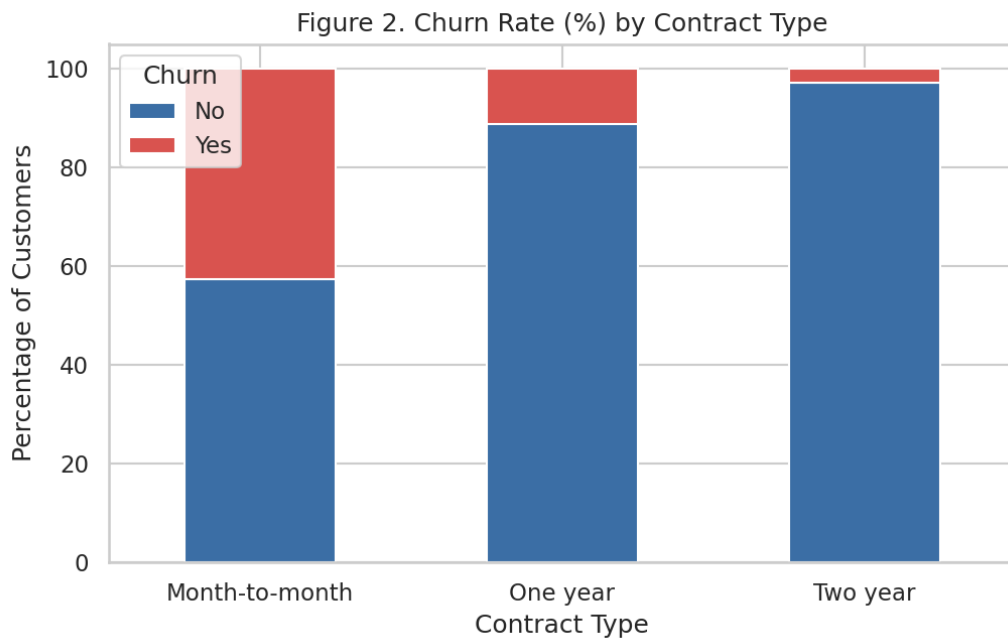


Figure 2. Churn Rate (%) by Contract Type

Contract type shows the starkest split of any categorical variable: 42.7% of month-to-month customers churn, versus 11.3% for one-year contracts and only 2.8% for two-year contracts. This is

the single most important behavioural signal in the dataset and strongly informed the choice to engineer TenureGroup and to prioritise Contract as a key predictor.

8.3 Bivariate / Relationship Analysis

```
sns.boxplot(data=df, x='Churn', y='MonthlyCharges')
sns.scatterplot(data=df, x='tenure', y='MonthlyCharges', hue='Churn')
```

Figure 3. Relationship Between Charges, Tenure and Churn

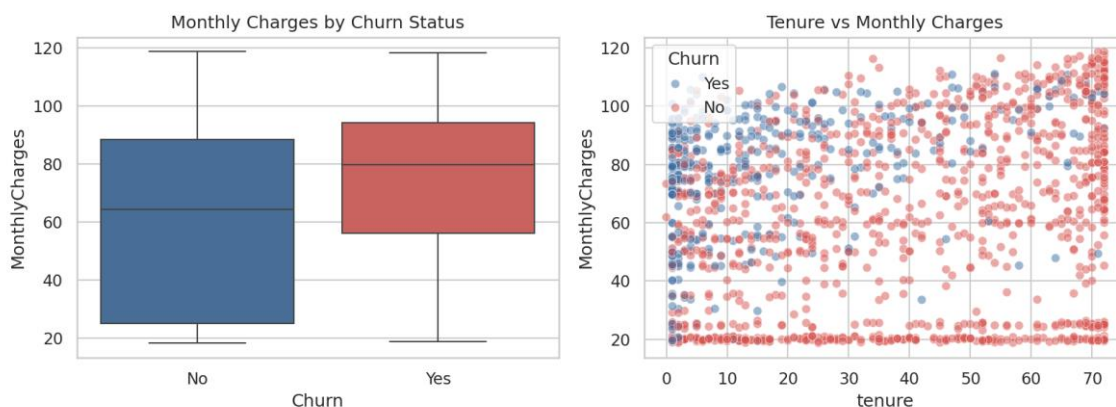


Figure 3. Relationship Between Charges, Tenure and Churn

Churned customers have a noticeably higher median MonthlyCharges than retained customers, and the scatter plot shows churn concentrated among low-tenure customers across most price levels, particularly above roughly \$70/month. This suggests price sensitivity is compounded by low tenure/loyalty rather than acting alone; the relationship is associative, not necessarily causal.

8.4 Correlation Analysis

```
sns.heatmap(df[num_cols].corr(), annot=True, cmap='coolwarm')
```

Figure 4. Correlation Heatmap of Numerical Variables

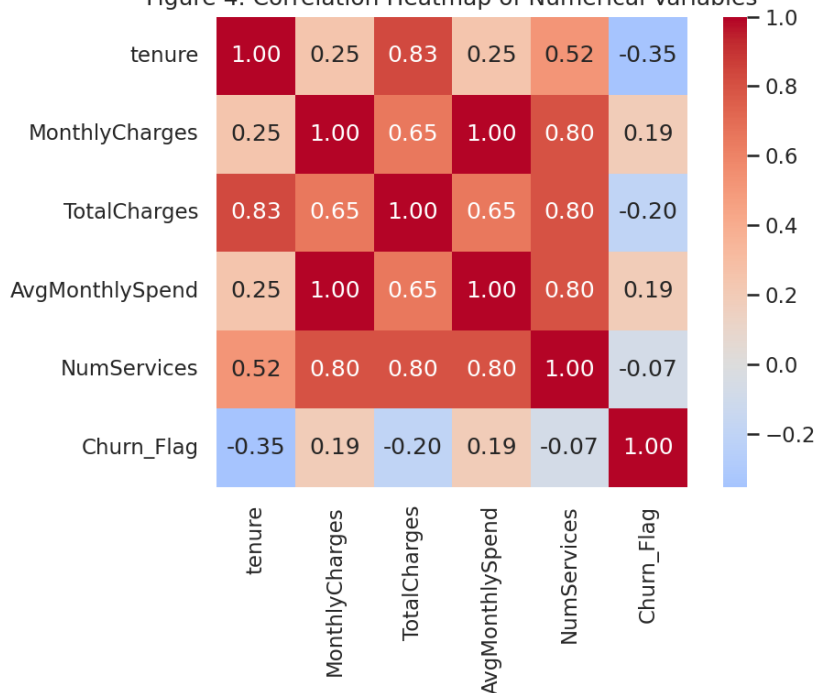


Figure 4. Correlation Heatmap of Numerical Variables

Churn_Flag correlates negatively with tenure ($r \approx -0.35$) and positively with MonthlyCharges ($r \approx 0.19$), consistent with the group comparisons above. TotalCharges correlates very strongly with tenure ($r \approx 0.83$), since it accumulates over the customer's lifetime; this collinearity was kept in mind when interpreting model coefficients and feature importances.

8.5 Other Figures

An additional chart on churn by internet service type is provided in Appendix B (Figure 6), showing that customers with fiber-optic internet churn at a substantially higher rate than DSL or no-internet customers.

9. STATISTICAL ANALYSIS

Variable	count	mean	std	min	25%	50%	75%	max	IQR
tenure	7043.0	32.37	24.56	0.0	9.0	29.0	55.0	72.0	46.0
MonthlyCharges	7043.0	64.76	30.09	18.25	35.5	70.35	89.85	118.75	54.35
TotalCharges	7043.0	2279.73	2266.79	0.0	398.55	1394.55	3786.6	8684.8	3388.05

Table 4. Descriptive Statistics of Key Numerical Variables

Average tenure is 32.4 months with a standard deviation of 24.6 months, reflecting the wide spread noted in Figure 1. MonthlyCharges has a mean of \$64.76 and a fairly symmetric spread (IQR of \$54.35), while TotalCharges is right-skewed (mean \$2,279.73 versus a median of \$1,394.55), as expected for an accumulating quantity across customers of very different tenure.

Churn	tenure	MonthlyCharges	TotalCharges
No	37.57	61.27	2549.91
Yes	17.98	74.44	1531.8

Table 4b. Mean Values by Churn Status

Customers who churned have, on average, less than half the tenure of retained customers (17.98 vs 37.57 months) and pay over \$13 more per month (\$74.44 vs \$61.27), despite having lower TotalCharges overall (a consequence of their shorter tenure, not lower pricing). This gap in group means directly supports the correlation results above and motivated including tenure, MonthlyCharges and their engineered derivatives in the feature set.

10. FEATURE ENGINEERING

Three additional features were engineered from the base variables to help the model and the analysis capture patterns that raw tenure or spend alone do not fully express:

Column 1 (Feature)	Column 2 (Definition)	Column 3 (Rationale)
TenureGroup	Bucketed version of tenure (0-1yr, 1-2yr, 2-4yr, 4+yr)	Simplifies non-linear tenure effect for grouped comparison
AvgMonthlySpend	TotalCharges / tenure (falls back to MonthlyCharges for new customers)	Normalises spend regardless of customer age

Column 1 (Feature)	Column 2 (Definition)	Column 3 (Rationale)
NumServices	Count of subscribed add-on/phone services (0-8)	Captures overall service engagement in one variable

Table 5. Engineered Features

11. MODEL DEVELOPMENT

11.1 Problem Formulation

This is a binary classification problem. The target variable is Churn_Flag (1 = the customer churned, 0 = the customer was retained). The input features are the 28 encoded demographic, service, contract, billing and engineered variables described in Sections 5-10. Classification is the correct formulation because the business decision (whether or not to target a customer for retention) is inherently a Yes/No outcome rather than a continuous quantity.

11.2 Model Selection

Two models were trained and compared: Logistic Regression, chosen as an interpretable baseline whose coefficients directly indicate the direction and relative strength of each feature's association with churn; and Random Forest, chosen because it can capture non-linear interactions (for example, between contract type and tenure) without requiring them to be manually specified. Both models are well suited to a moderately sized, mixed categorical/numerical tabular dataset like this one.

11.3 Training the Model

```
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

log_reg = LogisticRegression(max_iter=1000, random_state=42)
log_reg.fit(X_train_scaled, y_train)

rf = RandomForestClassifier(
    n_estimators=300, max_depth=8, min_samples_leaf=20, random_state=42
)
rf.fit(X_train_scaled, y_train)
```

Logistic Regression was trained with max_iter=1000 to ensure convergence on the encoded feature set. The Random Forest used 300 trees with max_depth=8 and min_samples_leaf=20 to limit overfitting on a dataset of this size, given that unconstrained trees can otherwise memorise the training data (see Section 12.3).

12. MODEL EVALUATION

12.1 Classification Evaluation

Figure 5. Confusion Matrix - Logistic Regression (Test Set)

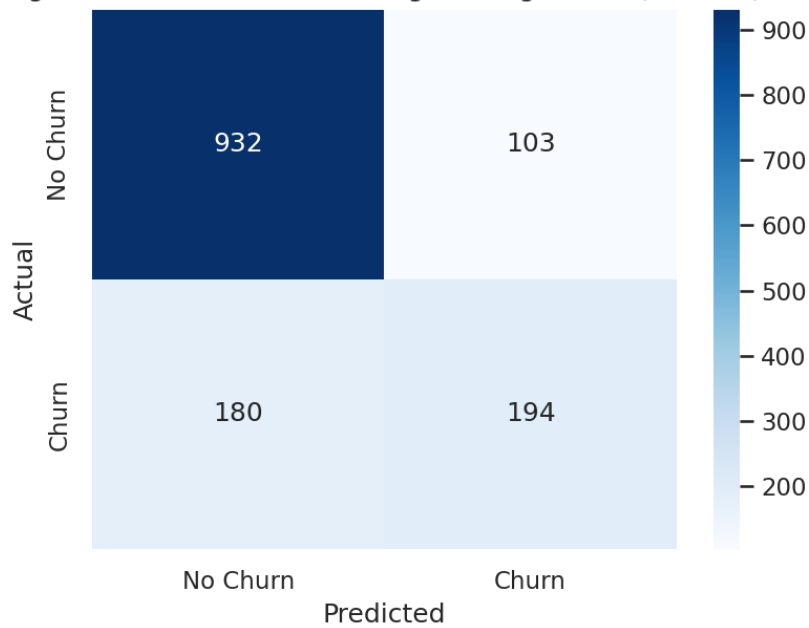


Figure 5. Confusion Matrix - Logistic Regression (Test Set)

Model	Train Accuracy	Test Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	81.0%	79.9%	65.3%	51.9%	57.8%	0.843
Random Forest	81.9%	79.8%	66.9%	47.6%	55.6%	0.847

Table 7. Classification Model Performance Comparison

Logistic Regression was selected as the primary model: it achieves the higher F1-score (57.8% vs 55.6%) and a comparable ROC-AUC (0.843 vs 0.847) to the Random Forest, while remaining more interpretable for a business stakeholder. On the test set, the model correctly identifies just over half of all customers who actually churn (recall = 51.9%), and when it flags a customer as likely to churn, it is correct about two-thirds of the time (precision = 65.3%). In this business context, a false negative (missing an at-risk customer) has a real revenue cost, while a false positive (offering a retention incentive to a customer who would have stayed anyway) is comparatively low-cost — which argues for monitoring recall closely, and potentially lowering the classification threshold below 0.5 in deployment to catch more at-risk customers, at the cost of some additional false positives.

12.2 Overfitting and Underfitting

Logistic Regression achieved 81.0% training accuracy versus 79.9% test accuracy — a gap of only 1.1 percentage points, indicating the model generalises well and is neither over- nor under-fitted. The Random Forest showed a slightly larger gap (81.9% train vs 79.8% test), a mild sign of overfitting that was already constrained by limiting tree depth and leaf size; a shallower or more heavily regularised forest, or additional training data, could narrow this gap further.

13. RESULTS AND FINDINGS

1. Contract type is the strongest behavioural driver of churn: month-to-month customers churn at 42.7%, versus 11.3% for one-year and 2.8% for two-year contracts (Figure 2), and Contract_Two year is among the top predictors in the Random Forest feature importance ranking (Figure 8).
2. Tenure is strongly protective against churn: churned customers average 17.98 months of tenure versus 37.57 months for retained customers (Table 4b), and tenure is both the top Random Forest feature (Figure 8) and negatively correlated with churn (Figure 4).
3. Higher monthly charges are associated with higher churn: churned customers pay \$74.44/month on average versus \$61.27 for retained customers (Table 4b, Figure 3), and MonthlyCharges/TotalCharges/InternetService_Fiber optic all rank among the top five most important features.
4. The Logistic Regression model predicts churn with 79.9% test accuracy, 65.3% precision, 51.9% recall and an F1-score of 57.8% (ROC-AUC 0.843), outperforming the Random Forest on F1-score while remaining more interpretable (Table 7, Figure 5).
5. Fiber-optic internet customers churn more than DSL or no-internet customers (Appendix B, Figure 6), and PaymentMethod_Electronic check is also a top-5 predictive feature, together suggesting service quality and payment friction as secondary churn drivers alongside contract and tenure.

14. DISCUSSION

The results consistently point to an early-tenure, month-to-month, higher-spend customer as the profile most likely to churn. This is intuitive: month-to-month contracts have no switching cost, so customers who are dissatisfied (with price, service quality, or otherwise) can and do leave immediately, whereas one- and two-year contracts create a natural retention floor. The fact that TotalCharges is highly correlated with tenure (Figure 4) but tenure itself is the stronger, more direct churn signal suggests that loyalty/relationship length matters more than cumulative spend per se. The moderate recall of the Logistic Regression model (51.9%) reflects the genuine difficulty of the problem: many of the input variables (demographics, most add-on services) are only weakly associated with churn, and important drivers such as customer service interactions, competitor offers, or life events are not captured in this dataset at all. It is also important to note that these are observed associations from historical data, not causal effects — for example, we cannot conclude from this analysis alone that moving a customer onto a longer contract will cause them to stay, only that customers on longer contracts have historically stayed at much higher rates.

15. LIMITATIONS

- Sample and scope: the dataset covers a single (anonymised) telecom operator at a single point in time; churn drivers may differ across markets, competitors, or time periods, limiting generalisability.
- Missing variables: the dataset does not include customer service interaction history, complaint records, competitor pricing, or macroeconomic factors, all of which plausibly influence churn.
- Class imbalance: only 26.5% of customers churned. This is not severe enough to require resampling techniques, but it does mean the model has more information about retained customers and contributes to the recall/precision trade-off discussed in Section 12.1.
- Correlation vs causation: the associations identified (e.g., contract type, tenure, charges) describe historical patterns, not proven causal mechanisms; a controlled experiment would be needed to confirm that changing a customer's contract or price would causally reduce their churn probability.

- Model scope: only two classical models were compared; more complex algorithms (gradient boosting, neural networks) or hyperparameter tuning via cross-validated grid/random search were out of scope for this training-stage capstone but could yield incremental performance gains.

16. CONCLUSION

This project set out to determine whether a customer's profile, services and billing history could predict subscription churn, and to identify the strongest drivers behind it. The analysis of 7,043 Telco customers shows that contract type, tenure and monthly charges are the dominant churn drivers, with month-to-month, early-tenure, higher-charge customers on fiber-optic plans forming the highest-risk segment. The Logistic Regression model built on these and 25 other features predicts churn with 79.9% test accuracy and a ROC-AUC of 0.843, correctly identifying roughly two-thirds of the customers it flags as at-risk. While the model is not accurate enough to fully automate retention decisions on its own, it provides a solid, interpretable basis for prioritising which customers a retention team should contact first, directly addressing the original business problem of allocating limited retention resources more effectively.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

- Prioritise retention outreach to month-to-month customers in their first 12 months, since this segment shows the highest observed churn rate (42.7%) and the lowest average tenure among churners (Table 4b, Figure 2); a proactive check-in or onboarding programme in months 1-3 could reduce early attrition.
- Offer incentivised upgrades from month-to-month to one- or two-year contracts (e.g., a modest discount or free add-on for committing to a longer term), since customers on longer contracts churn at 2.8-11.3% versus 42.7% for month-to-month customers.
- Review pricing or perceived value for high-monthly-charge customers, particularly fiber-optic subscribers, since both MonthlyCharges and InternetService_Fiber optic rank among the top predictors of churn (Figure 8) and churned customers pay \$13/month more on average than retained customers (Table 4b).
- Investigate service quality and payment experience for Electronic check users, since PaymentMethod_Electronic check is a top-5 predictive feature, suggesting either a correlated customer segment or friction in that payment channel worth investigating qualitatively.
- Deploy the Logistic Regression model as a monthly risk-scoring tool for the retention team, using a lowered probability threshold (below the default 0.5) to trade some precision for higher recall, given that missing an at-risk customer is more costly to the business than a low-cost retention offer extended to a customer who would have stayed regardless.

17.2 Future Work

Future iterations of this project could incorporate customer service and complaint logs, competitor pricing data, or usage patterns if available, to capture drivers not present in this dataset. Gradient boosting models (e.g., XGBoost or LightGBM) and hyperparameter tuning via cross-validation could be explored to improve recall without sacrificing precision. A controlled A/B test of the recommended retention interventions (contract upgrade incentives, targeted outreach) would help move from associative findings to causal evidence of what actually reduces churn.

18. REFERENCES

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<https://raw.githubusercontent.com/plotly/datasets/master/telco-customer-churn-by-IBM.csv>

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APPENDIX A. CODE

A.1 Data Loading and Cleaning

```
df = pd.read_csv(url)
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df['TotalCharges'] = df['TotalCharges'].fillna(0)
df = df.drop_duplicates(subset=['customerID'])
df['SeniorCitizen'] = df['SeniorCitizen'].map({0: 'No', 1: 'Yes'})
```

A.2 Preprocessing

```
model_df_encoded = pd.get_dummies(model_df, columns=categorical_cols, drop_first=True)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y)
scaler = StandardScaler()
X_train_scaled[num_cols] = scaler.fit_transform(X_train[num_cols])
X_test_scaled[num_cols] = scaler.transform(X_test[num_cols])
```

A.3 Model Training and Evaluation

```
log_reg = LogisticRegression(max_iter=1000, random_state=42)
log_reg.fit(X_train_scaled, y_train)
pred = log_reg.predict(X_test_scaled)
print(accuracy_score(y_test, pred), f1_score(y_test, pred))
```

The complete, executable code with all outputs is provided in the accompanying Jupyter/Colab notebook (Ain_Qurat_Capstone_Notebook.ipynb).

APPENDIX B. ADDITIONAL FIGURES AND TABLES

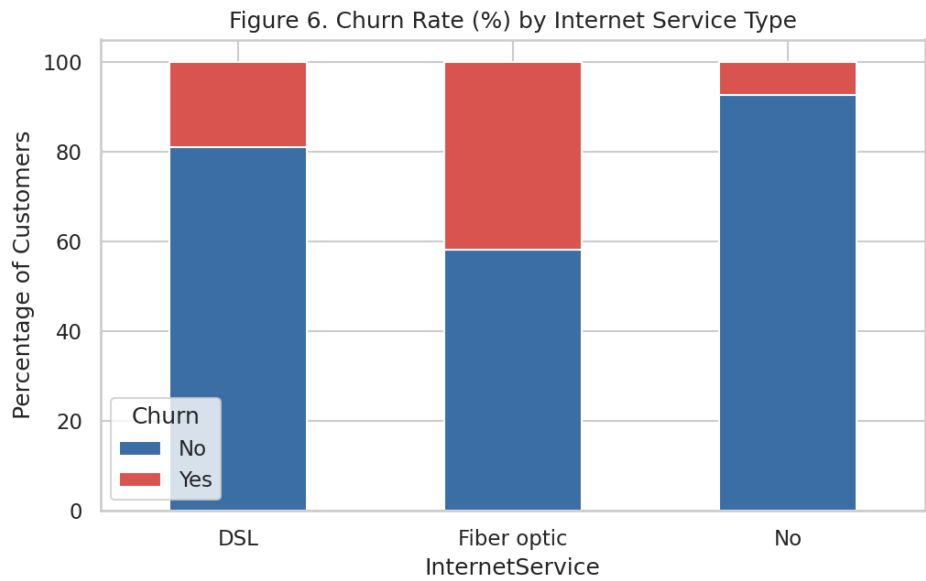


Figure 6. Churn Rate (%) by Internet Service Type

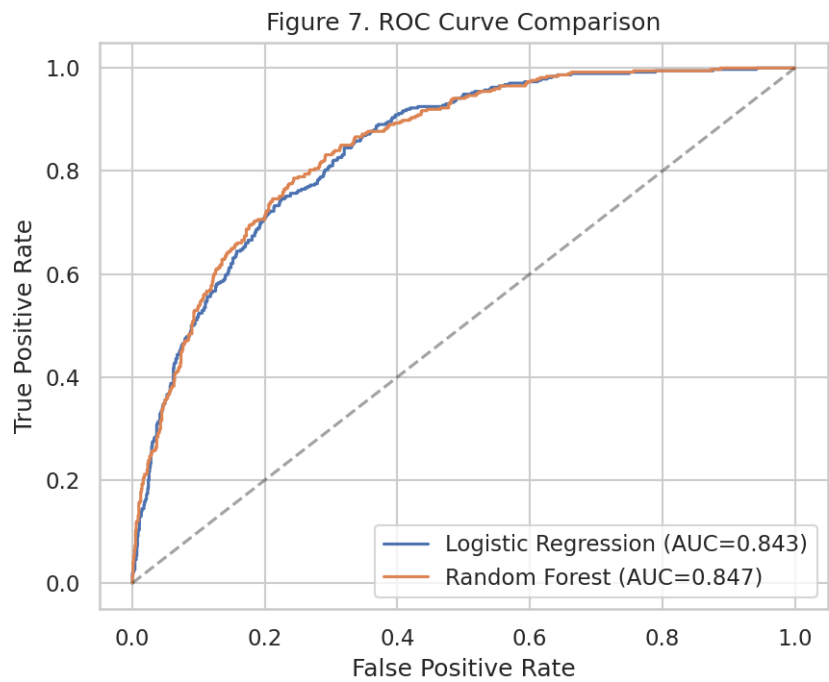


Figure 7. ROC Curve Comparison

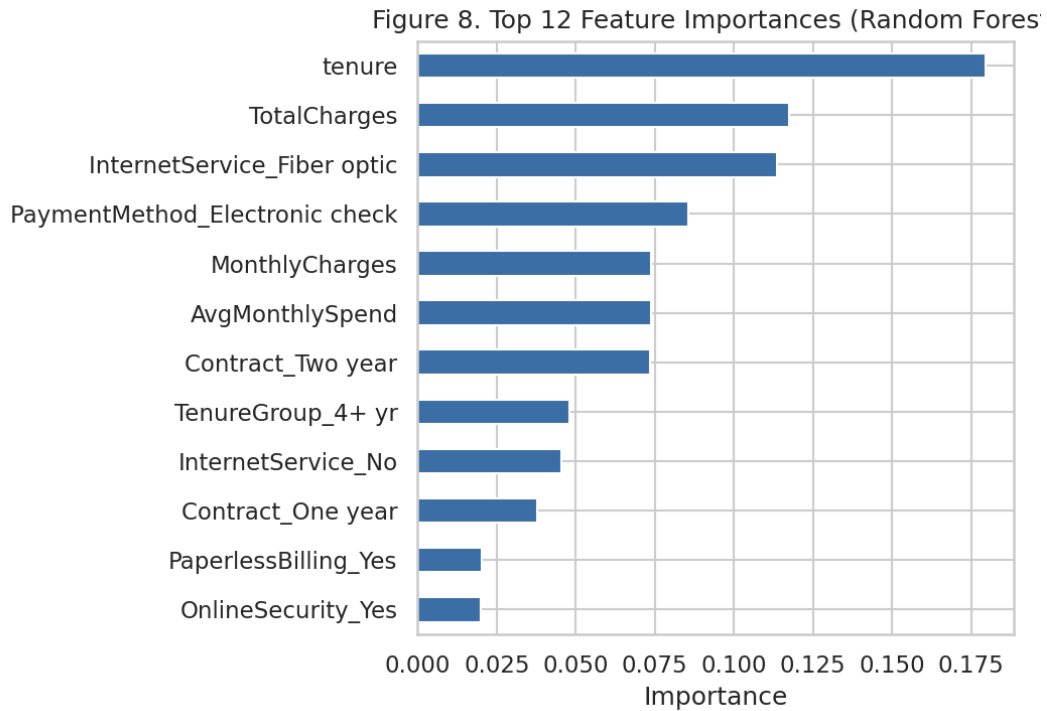


Figure 8. Top 12 Feature Importances (Random Forest)

Feature	Importance
tenure	0.1796
TotalCharges	0.1176
InternetService_Fiber optic	0.1136
PaymentMethod_Electronic check	0.0857
MonthlyCharges	0.0739
AvgMonthlySpend	0.0737
Contract_Two year	0.0734
TenureGroup_4+ yr	0.0481
InternetService_No	0.0456
Contract_One year	0.0377
PaperlessBilling_Yes	0.0202
OnlineSecurity_Yes	0.02

Table 8. Random Forest Feature Importance Ranking